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Vellore Institute of Technology
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CASE STUDY -1

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COURSE CODE: **CSI3005**

COURSE TITLE: **ADVANCED DATA VISUALIZATION TECHNIQUES**

FACULTY: **RAMESH C**

SLOT: **A1+TA1**

CLASS NUMBER: **VL2025260501536**

PROGRAMME: **M. TECH (INTEGRATED) - CSE**

SEMESTER: **WINTER 2025-26**

SCHOOL: **SCOPE**

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GitHub Repo Link: <https://github.com/abidhrx-cloud/Data-visualization-/blob/main/README.md>

[Q1] Dashboard Creation for a Given Dataset:

You have been assigned a unique dataset (available in Moodle). Your task is to design and develop an interactive dashboard using Tableau / Power BI / Shiny to effectively analyze the data and extract relevant business insights for decision-making.

Note: The completed work may also be uploaded to **GitHub**, and the repository link should be included in the submitted files.

Tasks:

1. **Data Exploration & Preparation**
 - Examine your dataset and identify key dimensions and measures.
 - Perform necessary preprocessing (handle missing values, outliers, and formatting).
2. **Dashboard Development**
 - Design and create a dashboard that includes:
 - Time-series analysis (Sales/Profit trends over time)
 - Category-wise and region-wise comparisons
 - Distribution of key variables
 - Relationship between variables (e.g., Sales vs Profit)
3. **Visualization Justification**
 - Select appropriate charts and provide justification for each visualization used.
4. **Interactivity**
 - Implement interactive features such as:
 - Filters (Date, Category, Region)
 - Drill-down capabilities
 - Tooltips
5. **Insight Generation**
 - Identify and report:
 - Key trends and patterns
 - Best and worst performing categories/regions
 - Any anomalies or unusual observations
6. **Performance Optimization**
 - Apply suitable techniques to handle large datasets efficiently.

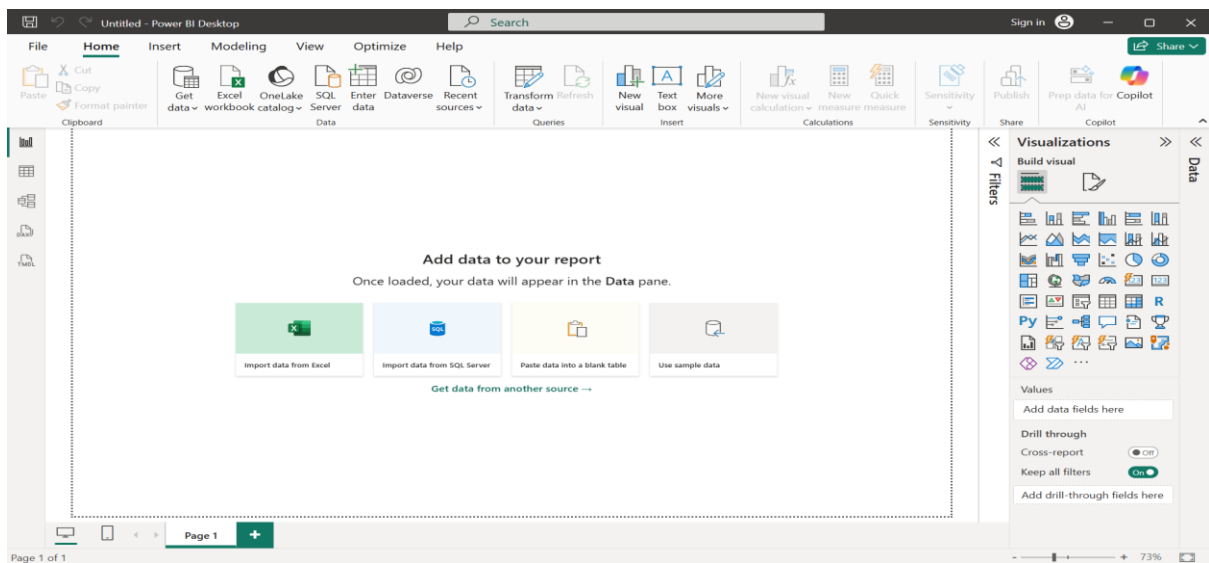
Dataset:cps91.csv

1) DATA EXPLORATION & PREPARATION

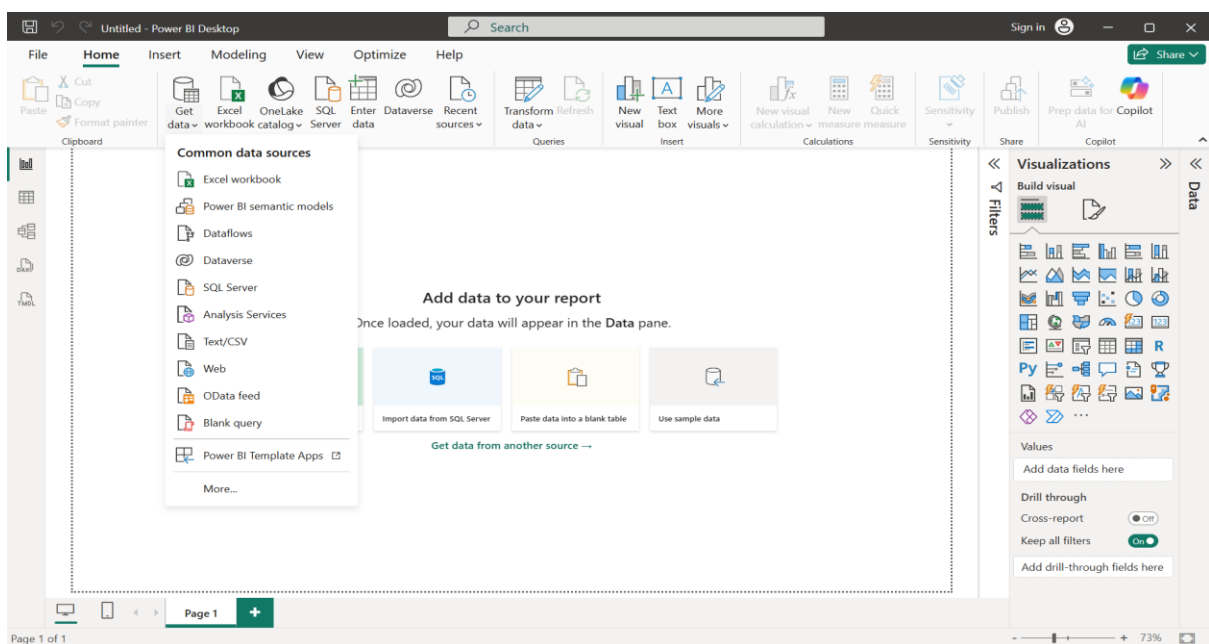
1.Data Loading

The dataset was imported into Microsoft Power BI using the Get Data option. The Transform Data option was selected to clean and prepare the data before analysis.

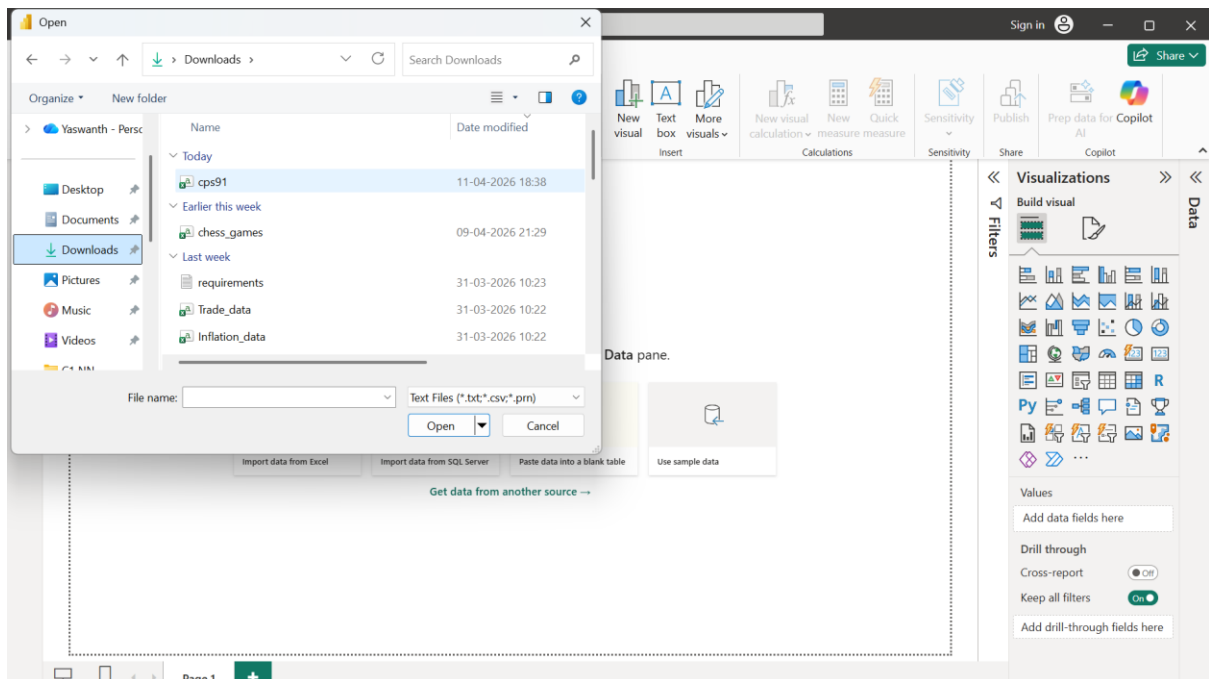
Open “PowerBI Desktop”:



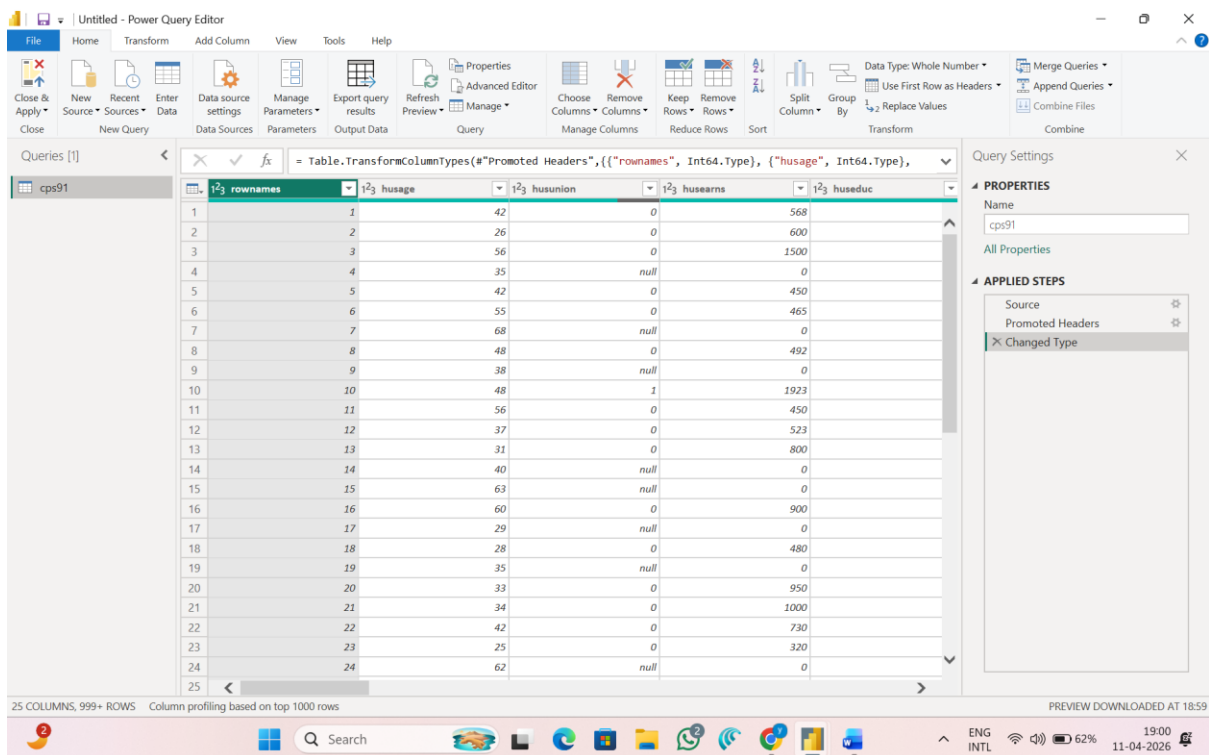
Click on “Get Data” and select respective file type of dataset:



Select the respective dataset:



Click on “Transform”:



Replacing null values:

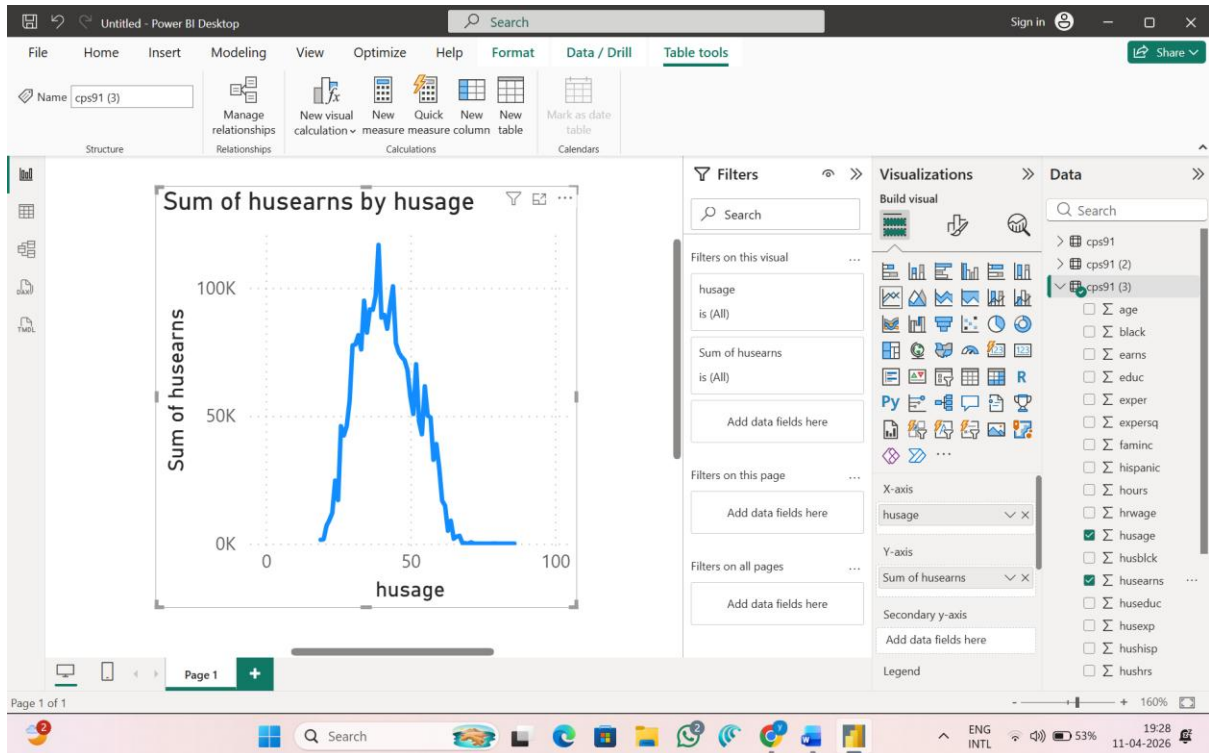
Power Query Editor interface showing the 'Replace Values' dialog box. The dialog prompts to replace one value with another in the selected columns. The 'Value To Find' is set to 'null' and the 'Replace With' is set to '0'. The background table shows columns: expersq, 1.2 nwifeinc, i23 inf, 1.2 hrwage, and 1.2 lwage. The formula bar displays: `= Table.ReplaceValue("#Replaced Value",null,0,Replacer.ReplaceValue,{"husunion"})`.

Deleting unnecessary columns:

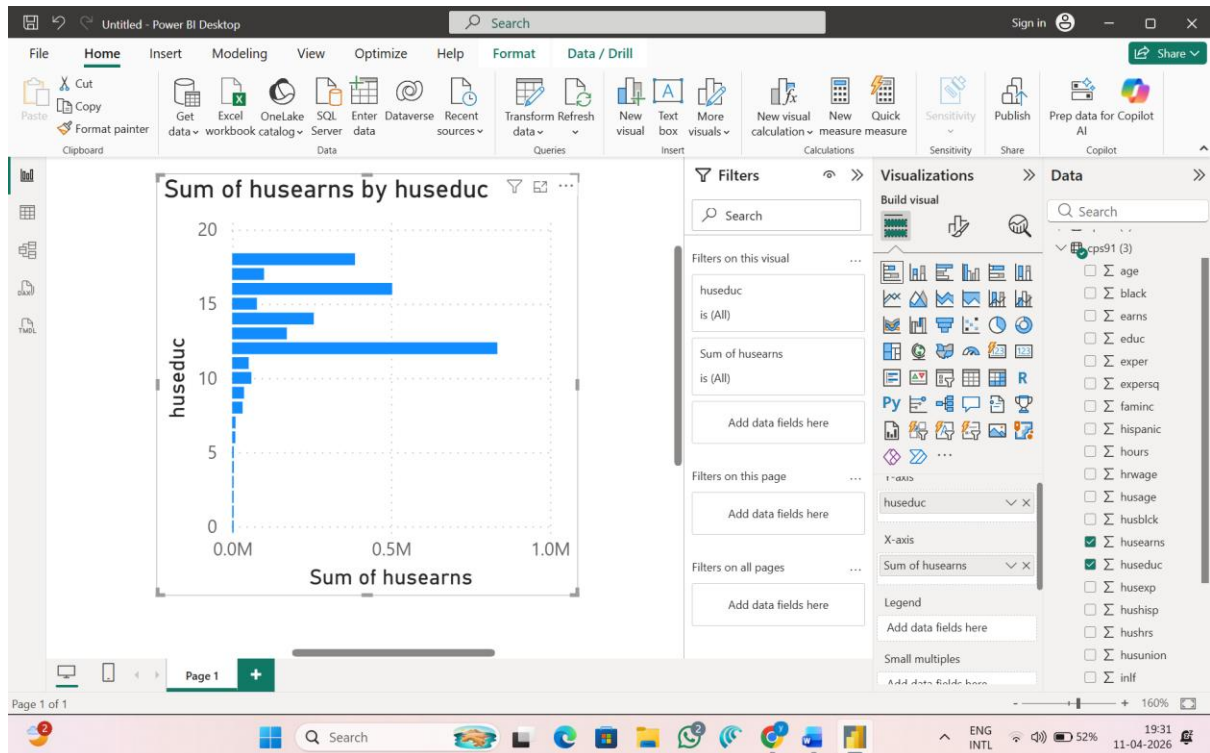
Power Query Editor interface showing the 'Delete column' dialog box. The dialog asks: "Are you sure you want to delete 'lwage'?". The background table shows columns: earns, age, black, educ, hispanic, union, faminc, husexp, exper, kidlt6, hours, expersq, nwifeinc, inf, and lwage. The 'Data' pane on the right lists the columns in the query.

2) DASHBOARD DEVELOPMENT:

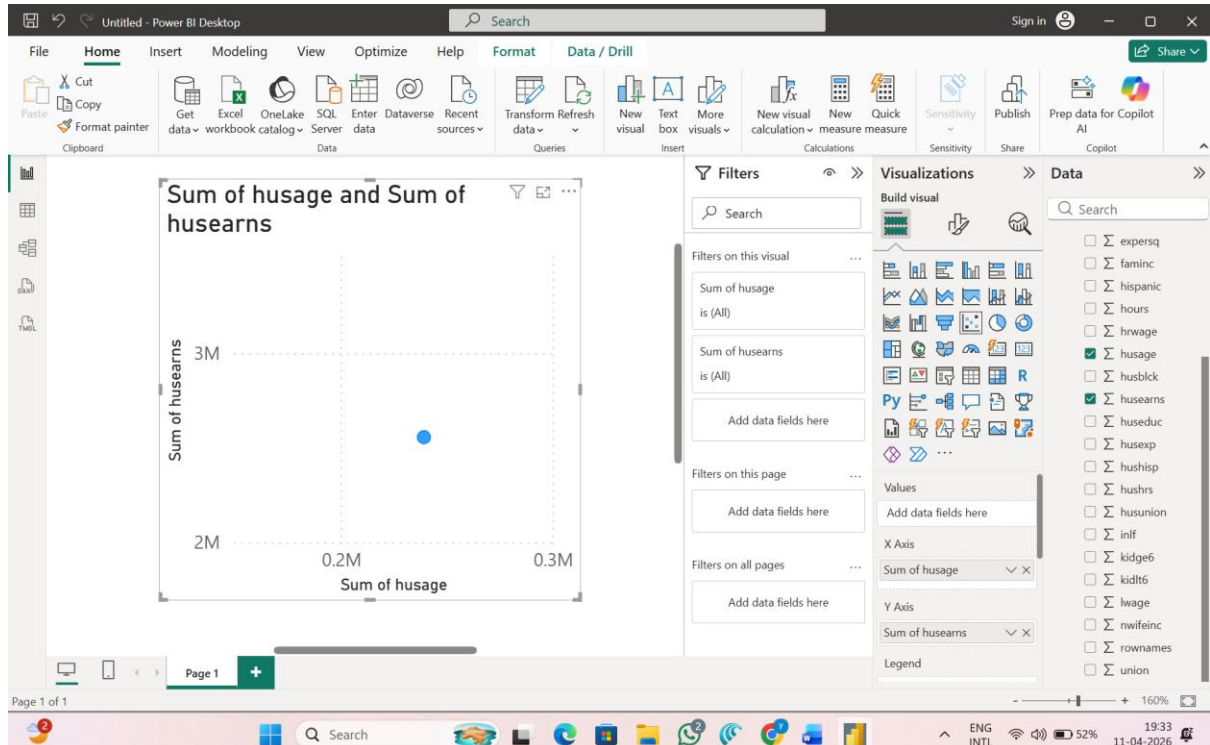
LINE CHART:



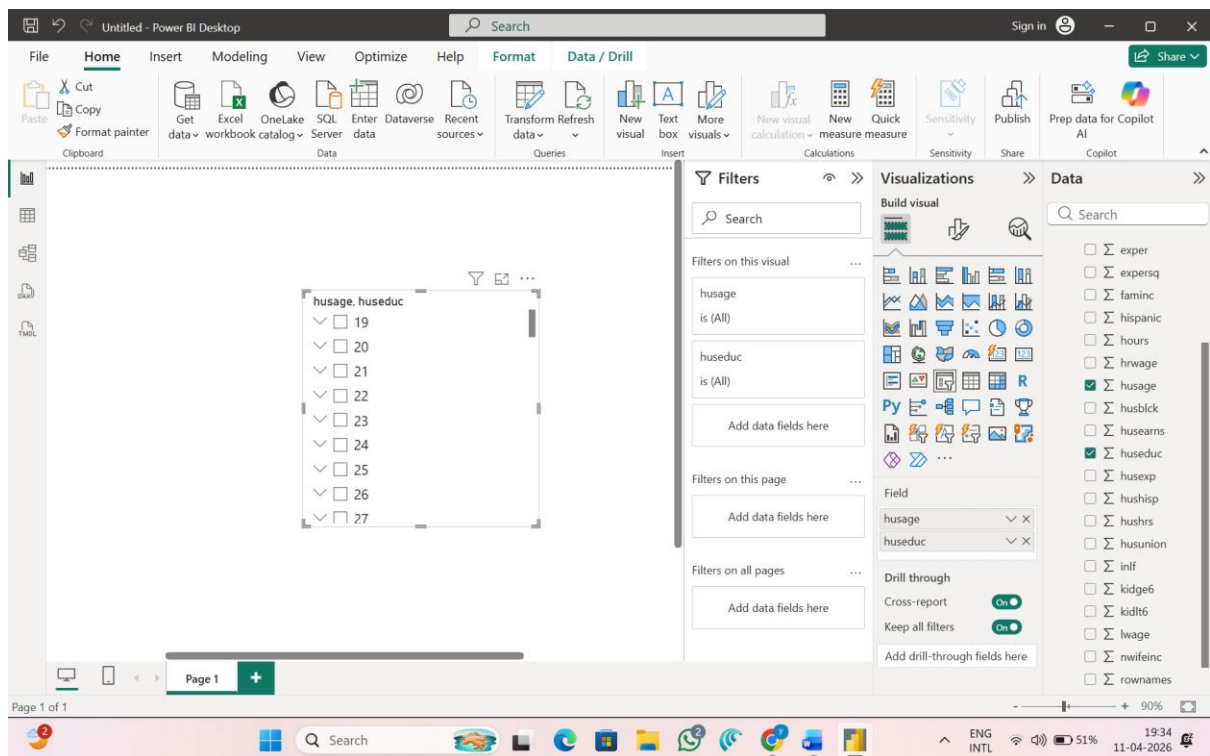
BAR CHART:



SCATTER PLOT:



ADDING A FILTER:



3. Visualization Justification

- Line chart is used to show trends between usage and earnings.
- Bar chart is used to compare earnings across education levels.
- Scatter plot is used to show the relationship between usage and earnings.

4. Interactivity

- Filters (slicers) are added to allow users to select specific data.
- Users can interact with the dashboard by selecting values in the filter.
- Tooltips provide additional information when hovering over charts.

5. Insight Generation

- There is a positive relationship between usage and earnings.
- Earnings differ across education levels.
- Some values are higher or lower, indicating variation in data.

6. Performance Optimization

- Unnecessary columns were removed.
- Missing values were handled.
- Only required visuals were used.
- Data was cleaned before creating the dashboard.

Insights

- There is a positive relationship between usage and earnings
- Earnings vary across education levels
- Some variations indicate possible outliers

Charts Used

- Line chart
- Bar chart
- Scatter plot

Interactivity

- Filters (slicers) are used for user interaction